

Galois extensions

Proposition (1) : L/k , M/k
finite

$$|\text{Emb}_k(M, L)| \leq [M:k]$$

Example:

$$M = \frac{\mathbb{Q}[x]}{(x^2+1)}$$

$$\text{Emb}_{\mathbb{Q}}(M, L) \xrightarrow{\sim} \{ \alpha \in L : \alpha^2 + 1 = 0 \}$$

$$\text{Emb}_{\mathbb{Q}}(M, \mathbb{R}) = \emptyset$$

$$\text{Emb}_{\mathbb{Q}}(\mathbb{M}, \mathbb{C}) \cong \{i, -i\}$$

$$\cdot \mathbb{Q}[x]/(x^2+1) \xrightarrow{\overline{\text{ev}_i}} \mathbb{C}$$

$$\begin{array}{ccc} x & \longmapsto & i \\ ax+b+(x^2+1) & \longmapsto & ai+b \end{array}$$

$$= \mathbb{Q}[x]/(x^2+1) \xrightarrow{\overline{\text{ev}_{-i}}} \mathbb{C}$$

$$\begin{array}{ccc} x & \longmapsto & -i \\ ax+b & \longmapsto & -ai+b \end{array}$$

Both embeddings have the same

image $\mathbb{Q}(i) = \{ai+b : a, b \in \mathbb{Q}\}$

(2) If we replace $\mathbb{Q}$ with $\mathbb{R}$
and $N = \mathbb{R}[x]/(x^2+1)$

$$\text{Emb}_{\mathbb{R}}(N, \mathbb{C}) \cong \{i, -i\}$$

$$\frac{\mathbb{R}[x]}{(x^4+1)} \xrightarrow{\quad \quad} \mathbb{C}$$

$$x \mapsto i$$

or

$$x \mapsto -i$$

Observation (1) If $M/k, L/k$ are

finite extⁿs with $[M:k] = [L:k]$,

then any $\lambda: M \rightarrow L$

$$\begin{array}{c} \downarrow \\ k \end{array} \quad \nearrow$$

in $\text{Emb}_k(M, L)$ is an isomorphism

Defⁿ $L/k; \text{Aut}(L/k)$

$$= \{ \lambda \in \text{Emb}_k(L, L) : \lambda \text{ an isomorphism} \}$$

The automorphism group of L/k

Observation (2)

If L/k is finite, then

$$\bullet \operatorname{Aut}(L/k) = \operatorname{Emb}_k(L, L)$$

Observation (3)

$\operatorname{Aut}(L/k)$ is a group

under composition

Corollary: L/k finite

$$|\operatorname{Aut}(L/k)| \leq [L:k]$$

Defⁿ L/k : finite extension

Then L/k is Galois

if $|Aut(L/k)| = [L:k]$

In this case, we will set

$$Gal(L/k) = Aut(L/k)$$

& call it the Galois group
for L/k

Examples:

$$\begin{array}{ccc} \textcircled{1} & \mathbb{M} = \mathbb{Q}[x] / (x^2 + 1) & \\ & \downarrow & \\ & \mathbb{Q} & \end{array} \quad x + (x^2 + 1) = \alpha$$

$$\text{Aut}(\mathbb{M}/\mathbb{Q}) = \text{Emb}_{\mathbb{Q}}(\mathbb{M}, \mathbb{M})$$

$$\begin{aligned} &\cong \left\{ \alpha \in \mathbb{M} : \alpha^2 + 1 = 0 \right\} \\ &= \{ \alpha, -\alpha \} \end{aligned}$$

$$\begin{array}{ccc} \alpha \mapsto \frac{\mathbb{Q}[x]}{(x^2 + 1)} & \longrightarrow & \frac{\mathbb{Q}[x]}{(x^2 + 1)} \\ \alpha & \longmapsto & \alpha \\ \text{i.e. } x & \longmapsto & x \end{array} \left. \vphantom{\begin{array}{ccc} \alpha \mapsto \frac{\mathbb{Q}[x]}{(x^2 + 1)} \\ \alpha & \longmapsto & \alpha \\ \text{i.e. } x & \longmapsto & x \end{array}} \right\} \text{identity}$$

$$\begin{array}{ccc}
 -d & \mapsto & \frac{(X^d - 1)}{(x^{d+1})} \longrightarrow \frac{(X^d - 1)}{(x^{d+1})} \\
 & & \alpha \mapsto -\alpha
 \end{array} \Bigg\} \tau$$

$$\begin{aligned}
 G \cong (M/\Phi) &= \{ \text{id}, \tau \} \\
 &\cong \mathbb{Z}/2\mathbb{Z}.
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad k &= \mathbb{F}_p, \quad M = \mathbb{F}_{p^d} \\
 [M:k] &= d
 \end{aligned}$$

$$\text{Fr} \in \text{Aut}(\mathbb{F}_{p^d}/\mathbb{F}_p)$$

↑

has order d

$$\langle Fr \rangle \leq \text{Aut}(\mathbb{F}_{p^d} / \mathbb{F}_r)$$

↑
has order d

↑
has order
at most d

$\Rightarrow \mathbb{F}_{p^d} / \mathbb{F}_r$ is Galois

$$\& \text{Gal}(\mathbb{F}_{p^d} / \mathbb{F}_r) = \langle Fr \rangle$$

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$$\mathbb{Z}/d\mathbb{Z}.$$

$$(3) \quad M = \frac{\mathbb{Q}[x]}{(x^3 - 2)}$$

$$\begin{aligned} \text{Emb}_{\mathbb{Q}}(M, M) &\cong \{ \alpha \in M : \alpha^3 = 2 \} \\ &= \{ x + (x^3 - 2) \} \end{aligned}$$

$$\Rightarrow | \text{Aut}(M/\mathbb{Q}) | = 1.$$

$M/\mathbb{Q}$ is not Galois